Assignment 10: Linked List Assignment

1. **Write a method that allows the user to insert a new node at any specified position within a Singly Linked List.**

   Note: - the counting of nodes here starts from 1 not 0.

   Test Data:
   Original List: 7 5 3 1
   Insert a node with value 12 at position 1
   Expected Output List: 12 7 5 3 1

   Original List: 12 7 5 3 1
   Insert a node with value 14 at position 4
   Expected Output List: 12 7 5 14 3 1

   Original List: 12 7 5 14 3 1
   Insert a node with value 18 at position 7
   Expected Output List: 12 7 5 14 3 1 18

2. **Write a method that deletes the n^{th} node from the end of a Singly Linked List.**

   Note: - The deletion happens from the end of the linked list. In other words, if we want to delete the 1st node, it will be the last node (since we count from the end, not from the start).

   Test Data:
   Original List: 7 5 3 1
   Delete the 2nd node from the end
   Expected Output List: 7 5 1

   Original List: 7 5 1
   Delete the 3rd node from the end
   Expected Output List: 5 1

3. **Write a method that returns the value of the n^{th} node in a given Singly Linked List.**

   Test Data:
   Original List: 7 5 3 1
   Position: 1
   Expected Value: 7

   Position: 2
   Expected Value: 5

   Position: 3
   Expected Value: 3

   Position: 4
   Expected Value: 1

4. **Write a method that inserts a new node at the middle position of a given Doubly Linked List.**

   Test Data:
   Original Doubly Linked List:
   Forward Traversal: Orange White Green Red
   Reverse Traversal: Red Green White Orange

   Insert a node with value Pink at the middle position
   Expected Forward Traversal: Orange White Pink Green Red
   Expected Reverse Traversal: Red Green Pink White Orange

   Insert a node with value Black at the middle position
   Expected Forward Traversal: Orange White Pink Black Green Red
   Expected Reverse Traversal: Red Green Black Pink White Orange